

# Introductory Econometrics with Recitation — Quiz 1

March 10, 2026

1. **(10 points)** Let  $X$  be a random variable such that  $E(X) = 6$  and  $\text{Var}(X) = 9$ . Let  $Y = -4X + 10$ .
  - (a) **(5 points)** Find  $E(2X + Y)$ .
  - (b) **(5 points)** Find  $\text{Var}(3Y)$ .

2. **(20 points)** Data on monthly customer satisfaction index scores (measured on a 0-500 scale) for  $n = 360$  retail branches in Brussels yield an average score of  $\bar{Y} = 432.6$  and a standard deviation of  $s_Y = 19.8$ .

Suppose the branches are divided into those with small sales teams ( $< 20$  employees) and large sales teams ( $\geq 20$  employees). The following results are found:

| Team Size | Average Score ( $\bar{Y}$ ) | Standard Deviation ( $s_Y$ ) | $n$ |
|-----------|-----------------------------|------------------------------|-----|
| Small     | 439.7                       | 20.6                         | 140 |
| Large     | 428.1                       | 17.9                         | 220 |

We want to know if there exists statistically significant evidence that the average customer satisfaction scores differ between branches with small and large sales teams. Assume all relevant conditions are met.

- (a) **(3 points)** Write the hypotheses in symbols and in words.
- (b) **(6 points)** Calculate the test statistic  $T$  and the associated degrees of freedom.
- (c) **(4 points)** Find the p-value and interpret it in the context.
- (d) **(2 points)** What is the conclusion of the hypothesis test at  $\alpha = 0.05$ ?
- (e) **(5 points)** Construct a 90% confidence interval for the mean customer satisfaction score in the population of all branches.